

CONTACT

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SUMMARY

I specialize in firmware architecture for resource-constrained MCUs, focusing on power efficiency and IoT connectivity. With hands-on experience in Bluetooth (BLE & Mesh), RTOS, and peripheral driver development, I design modular, scalable, and power-efficient firmware for embedded systems. Passionate about innovation and problem-solving, I develop high-performance solutions that drive the next generation of smart technologies.

- SKILL
- **Programming Languages:** C, Python
  - **Microcontrollers & SoCs:** Nordic nRF Series, TI MSP, STM32, PIC, ESP32, Arduino
  - **EmbeddedOS & RTOS:** ZephyrOS, Bare-metal
  - **Wireless Protocols:** Bluetooth Low Energy, Bluetooth Mesh
  - **Communication Protocols:** UART, SPI, I2C, MQTT
  - **Tools & Debugging:** J-Link, Logic Analyzers, VSCode, Github
  - **Cloud & IoT:** Azure Cloud
  - **Firmware & System Development:** Bootloader, Low-Power Optimization

- PROFESSIONAL EXPERIENCE
- Firmware Engineer | Oct 2024 - Present**  
Austin Circuit Design, Texas

- Designed and proposed tag tracking solutions using Nordic nRF52/nRF53 gateways and Atmosic ATM3330e tags for asset tracking.
  - Developed BLE & Bluetooth Mesh firmware and integrated sub-GHz RFM69 (SPI) and Ambiq AM1805 RTC (I2C) drivers.
  - Implemented Azure IoT MQTT connectivity via ESP32 (Wi-Fi), interfacing with gateways over UART.
  - Optimized power consumption across multiple modules, achieving designed power limits and with custom communication strategies for minimum up time.

**Firmware Engineer Intern | Jul 2024 - Sep 2024**  
Austin Circuit Design, Texas

- Developed precision weight scale firmware with signal processing, calibration, and real-time data acquisition using TI MSPMOC1104 and STM32F103 MCUs.
  - Designed and implemented bootloader functionality for firmware updates.
  - Created UART command-line interface (CLI) in Python for computer-based communication with Arduino.

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**PROJECTS**

**Capstone Project | Design and Implement of Self balancing – Line following Motorcycle**

Bach Khoa Ho Chi Minh city University of Technology | Jan 2024 – May 2023

- Developed MPU6050 driver with Kalman filter for accurate motion sensing.
- Implemented motor controllers for self-balancing, steering, and line-following on Arduino Mega.
- Integrated ESP32-CAM for Wi-Fi-based line tracking & computer communication.
- Designed image processing algorithm with OpenCV for line detection.

**Mechatronics Projects | Line Tracking and Color Identifying Mobile Robot**

Bach Khoa Ho Chi Minh city University of Technology | Aug 2023 – Dec 2023

- Led the project, designing mechanical, electrical & controller solutions.
- Developed PID motor & Lyapunov-based line tracking controllers in MATLAB.
- Implemented color-based turn decision algorithm on Arduino Nano.
- Optimized system performance through real-world testing.

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**EDUCATION &  
CERTIFICATIONS**

**Bach Khoa Ho Chi Minh city University of Technology**

Bachelor of Engineering – Mechatronics Engineering

Sep 2020 – Jun 2024

- GPA: 8.09/10
- English Program

**English – IELTS ACADEMIC BAND 7.0**

Issued in Aug 2020

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**HONOR & AWARDS**

- Academic Incentive Scholarship (100% tuition fee) – Semester 221, 231, 232
- Participation Certification – The EAI International Conference on Renewable Energy and Sustainable Manufacturing – 2024